

Quality Assurance & Test Cases Report

Project: Vaultory — Small Business Inventory and Sales App (SBISA)

Formal Test Verification & Acceptance Deliverable (Sprint 3 / VAU-029 / SOW D-8)

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Client / Sponsor	Small Business Retailer (Prof)	Project Manager	Laxman Patel
Base Documents	BRD v3.4 · SRS v1.2 · SOW v1.2 · Sprint Planner v3.0		

Revision History

Version	Date	Authors	Description of Changes
1.0	22/09/2026	Rohan Vashisht & Anoop Gupta	Initial formal test verification report covering all Acceptance Criteria (T-AC1–14) and functional modules.

Approvals & Sign-Off Baseline

1 Executive Summary for the Client

This **Quality Assurance & Test Cases Verification Report** presents an authoritative, client-oriented summary of the quality testing conducted on the **Vaultory** multi-store retail platform. Developed under Agile Scrum across 3 dedicated sprints (29 August – 30 September 2026), Vaultory delivers an integrated web platform for multi-location inventory tracking, point-of-sale operations, safety stock alerting, AI-powered automated procurement, and executive business analytics for a retail business with **3 physical branches** (Store A – MG Road, Store B – Indiranagar, Store C – Jayanagar) and a **Central Warehouse**.

Role / Designation	Name	Governance Status	Date
Client / Sponsor	Prof	Pending Formal UAT Review	—
Project Manager	Laxman Patel	Scope & Acceptance Verified	22/09/2026
Business Analyst	Ved Naik	Requirements Baseline Traced	22/09/2026
Solutions Architect	Anoop Gupta	Technical Integrity Validated	22/09/2026
Scrum Master	Devdarshan S	Sprint 3 QA Exit Certified	22/09/2026
Tech Lead	Rohan Vashisht	Test Suites & Code Certified	22/09/2026

Key Stakeholder Highlights & Confidence Statement

- **100% Requirements Verification:** All 14 formal Acceptance Criteria (T-AC1 to T-AC14) from SRS Section 13 and all 10 Business Use Cases (UC-01 to UC-10) have passed testing.
- **Comprehensive Automated Test Suite:** A total of 208+ automated tests across 19 dedicated suites (12 backend unit stores, 7 frontend reactive and schema suites) run with zero failures.
- **Strict Negative Stock Prevention:** Atomic database transaction locks (fn_mutate_stock) enforce zero negative stock across all sales, returns, and inter-store transfers.
- **Commercial Data Masking:** Proprietary supplier purchase prices and wholesale costs are masked (0.00) to frontline cashiers, sales staff, and store managers, protecting trade secrets.
- **Fault-Tolerant AI Architecture:** The AI reordering and warehouse replenishment module integrates with high-speed Groq inference models with an automatic, instantaneous fallback to deterministic Simple Moving Averages (SMA) should cloud connectivity ever drop.
- **Zero Critical Defects:** The system operates cleanly on its production cloud architecture (Vercel + Render + Supabase), with page loads averaging 0.38 seconds (exceeding the 3.0-second SLA).

2 Quality Assurance Scope & Methodology

Testing followed a multi-tiered quality assurance strategy ensuring thorough verification across all operational layers:

1. **Tier 1: Core Business Rule Testing (Unit Tier):** Validates standalone calculations, tax and discount logic, SKU prefix generation, safety stock threshold validations, and role-based data masking.
2. **Tier 2: Cross-Module Integration Testing (Integration Tier):** Validates end-to-end data flows where a point-of-sale transaction decrements inventory atomically, writes an audit record, updates daily ledger views, and trips low-stock alerts.

3. **Tier 3: User Journey & Acceptance Verification (E2E Tier):** Validates realistic cashier, store manager, and executive workflows across multiple retail stores, goods-in delivery processing, and CSV report downloads.

3 Acceptance Criteria Verification Matrix (SRS & BRD)

The table below maps every formal acceptance criterion from **SRS Section 13** to its verified outcome:

4 Core Business Use Cases Verification (UC-01 to UC-10)

Each of the 10 core operational use cases defined in **SRS Section 12** was walked through end-to-end:

5 Module-by-Module Test Case Breakdown

5.1 Authentication, Session Security & Access Control

- **Credential Verification:** Validates standard email/password authentication, issuing signed JWT tokens with persistent session restore across browser reloads.
- **One-Time Password (OTP):** Validates 6-digit email OTP magic-link login flows for secure, passwordless authentication.
- **Public Signup Role Clamp:** Self-service signups are clamped server-side to the base `store_staff` role with a required physical store assignment. Elevation to Administrator requires explicit administrative assignment.
- **Session Teardown:** Signing out clears authorization tokens immediately and prevents access to protected application routes.

5.2 User Administration & Branch Scoping

- **Role Permissions:** Enforces authorization matrix across 5 roles: `admin`, `senior_stakeholder`, `store_manager`, `store_staff`, and `sales_personnel`.
- **Multi-Store Isolation:** Store personnel assigned to Store A cannot view, modify, or deduct inventory belonging to Store B or Store C.
- **Reversible Soft Archival:** Employee deactivation suspends login privileges while preserving sales receipts and audit history.

5.3 Product Catalog, Categories & Master Units

- **Automated SKU Generation:** Intelligently derives standard SKU prefixes from category names (e.g., “Mobile Accessories” → PMA-001, “Electronics” → PELEC-005).
- **Hierarchical Categories:** Supports parent-child category structures while preventing circular parent loops.
- **Unit Configuration:** Handles diverse measurement units (pieces, kilograms, boxes, cartons, liters) with abbreviation validation.

5.4 Multi-Location Inventory & Stock Movements

- **Atomic Database Transactions:** Deductions and additions utilize atomic stored procedures (`fn_mutate_stock`), ensuring concurrent POS checkouts never corrupt stock registers.
- **Zero Negative Stock Enforcement:** Transactions attempting to reduce inventory below zero are rejected with user-friendly error dialogs.

- **Immutable Audit Ledger:** Every inventory mutation records an unalterable log containing timestamp, user ID, location, quantity delta, and business reason.

5.5 Point of Sale (POS), Cashier Operations, Voids & Returns

- **Rapid Checkout:** POS counter sales immediately decrement stock, generate sequential receipts (SAL-2026-XXXX), and update daily revenue summaries.
- **Overselling Prevention:** POS halts checkout and alerts cashier if requested item quantity exceeds shelf availability.
- **Controlled Customer Returns:** Customers can return items from earlier purchases; system validates returned quantities against original sales and restores items to store shelves.
- **Transaction Voiding:** Authorizing managers can void mistaken transactions with a mandatory audit note, restoring unreturned merchandise and excluding the sale from revenue figures.

5.6 Sales Analytics, Reporting & Data Export

- **Daily Sales Ledger:** Aggregates gross revenue, discounts, returns, net sales, and units sold, with store and product filtering.
- **Quarterly & Annual Trends:** Consolidates sales volume across monthly and quarterly periods for corporate tax and executive reviews.
- **Cross-Store Comparison:** Ranks branch performance across Store A, Store B, and Store C.
- **Data Export Utility:** Generates cleanly formatted, formula-sanitized CSV files and print layouts.

5.7 Safety Stock Rules, Reorder Thresholds & Alert Center

- **Logical Threshold Verification:** Enforces business rule: $\text{Safety Stock} \leq \text{Reorder Point} \leq \text{Target Stock Level}$.
- **Automated Alert Triggers:** Real-time creation of LOW_STOCK warnings when stock drops to or below the reorder point, and OUT_OF_STOCK alerts when stock reaches zero.
- **Notification Preferences:** Staff can configure personal alert preferences, mark notices as read, and track restock resolutions.

5.8 Supplier Directory & Purchase Order (Procurement) Lifecycle

- **Vendor Directory:** Manages vendor details, email contacts, and contractual delivery lead times (default: 7 days).
- **Five-Stage Order Lifecycle:** Tracks orders seamlessly through `draft` → `sent` → `partially_received` → `received` → `closed`.
- **Duplicate Order Protection:** Rejects unintentional duplicate purchase orders for identical products and stores.
- **Partial Delivery Receipts:** Supports partial shipments (e.g., receiving 50 of 100 ordered units), updating on-hand stock while keeping the remaining balance active.
- **Perishable Goods Expiry:** Mandates expiry date capture when receiving perishable food and grocery deliveries.

5.9 AI Demand Forecasting & Warehouse Replenishment

- **Deterministic Baseline (SMA):** Calculates historical moving averages across 30-day sales windows to compute mathematical demand projections.

- **Groq AI Advisory:** Generates natural-language demand rationales based on recent sales trends.
- **Fault-Tolerant Fallback:** In the event of an external AI service disruption or missing API credentials, the system automatically falls back to deterministic SMA calculations with zero system interruption.
- **Replenishment Workflow:** Evaluates combined sales across all 3 stores to advise Central Warehouse stock; managers can Accept (generating a draft PO), Modify, or Reject.

5.10 Commercial Confidentiality & Cost Price Masking

- **Field-Level Masking:** Wholesale product cost prices are masked (\$0.00) for cashiers and store staff.
- **API Security:** Sensitive financial figures are sanitized at the server level, preventing technical inspection via browser developer tools.

5.11 Frontend User Interface, Validation & Usability

- **Client-Side Validation:** Real-time Zod schemas prevent invalid submissions (e.g., negative prices, empty item names).
- **Reactive Data Sync:** TanStack Query caches data and performs background updates, eliminating slow full-page reloads.
- **Responsive UI:** Fully responsive across desktop, laptop, and tablet screens with accessible contrast and touch-friendly controls.

6 Defect Prevention, Safeguards & Edge Case Testing

7 System Performance & Non-Functional Verification

8 Conclusion & Handover Recommendation

Final Quality Assurance Certification

The engineering team formally certifies that **Vaultory** has successfully passed all quality gates for **Sprint 3 Deliverable VAU-029 (Full Test Pass)**:

- **100% Requirements Compliance:** All 14 Acceptance Criteria and 10 Use Cases are fully satisfied.
- **Automated Test Coverage:** 208+ automated tests pass with 0 defects remaining.
- **Business Stability:** Zero negative stock, robust multi-store isolation, cost price confidentiality, and resilient AI forecasting are operational.

Recommendation: Vaultory is certified as **Production Ready** and recommended for formal **Client User Acceptance Testing (UAT) and Demonstration** (Deliverable D-8).

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Test ID	Acceptance Description	Expected Business Result	Verified Outcome	Status
T-AC1	Multi-Location Stock Visibility	Accurate stock counts across 3 stores and warehouse within 3.0s.	Real-time stock counts load in 0.38s; status badges display cleanly.	✓ PASSED
T-AC2	Stock Operations & Integrity	In/Out/Transfer/Adjust update atomically; negative stock is blocked.	Atomic mutations guarantee ledger consistency; negative stock blocked.	✓ PASSED
T-AC3	Sales Recording & Reports	Sale decrements stock; invoice created; reports update immediately.	Real-time deduction; SAL-YYYY-XXXX issued; daily views sync.	✓ PASSED
T-AC4	Safety Stock & Alerts	Low-stock alert when balance $\leq$ reorder; out-of-stock when 0.	Threshold evaluations raise notification; alerts can be resolved.	✓ PASSED
T-AC5	Automated AI Reordering	Auto-PO generated for items below reorder point with correct quantity.	Reorder scan groups by vendor; drafts POs without duplicates.	✓ PASSED
T-AC6	PO & Goods-In Lifecycle	Full PO lifecycle (Draft $\rightarrow$ Closed); receiving increases stock.	Partial and full goods-in verified; expiry date enforced for perishables.	✓ PASSED
T-AC7	AI Warehouse Advice	AI advises central warehouse target levels with transparent rationale.	Rationale displayed; Accept/Modify/Reject workflow logged.	✓ PASSED
T-AC8	Store Sales Performance	Staff and managers view branch sales and cross-store comparisons.	Store-filtered sales, unit counts, and comparative charts verified.	✓ PASSED
T-AC9	Executive KPI Dashboard	High-level inventory value, revenue, and turnover drill-downs.	Consolidated KPIs load in 0.42s; figures reconcile with cash registers.	✓ PASSED
T-AC10	Authentication & RBAC	Email/password and OTP login; strict role and store permission guards.	Self-signup locked to <code>store_staff</code> ; unauthorized calls return 403.	✓ PASSED
T-AC11	Cost Price Data Masking	Wholesale supplier prices masked to non-admin / non-stakeholder roles.	Stripped at backend API level; cashiers see \$0.00 cost price.	✓ PASSED
T-AC12	Cloud Hosting & Architecture	Stable operation on Vercel + Render + Supabase free-tier stack.	100% uptime during test runs; database pooling verified stable.	✓ PASSED
T-AC13	Performance & Latency	Interactive views $\leq$ 3.0s p95; reports sync within 5 minutes.	Latency under 450ms p95; database views refresh instantaneously.	✓ PASSED
T-AC14	Scope Integrity	Verified strict adherence to approved BRD v3.4 scope.	Zero out-of-scope feature creep; clean audit trail preserved.	✓ PASSED

Use Case	Operational Flow	Primary Actor	Verified Operational Behavior	Result
UC-01	Record Counter Sale	Sales Personnel	Cashier adds items, applies discount, confirms sale. Stock decrements immediately, invoice generated, recorded in ledger.	✓ VERIFIED
UC-01a	Void Sale / Return	Store Staff / Admin	Void: Admin gives mandatory reason; stock restored, sale voided. Return: Line items returned; refund computed.	✓ VERIFIED
UC-02	Stock-In / Receive Goods	Store Staff / Admin	Delivery received against open PO; quantities verified; perishable expiry dates logged; shelf inventory updated.	✓ VERIFIED
UC-03	Auto-Reorder at Threshold	System / Admin	Inventory scan detects items $\leq$ reorder point; drafts grouped purchase orders by vendor automatically.	✓ VERIFIED
UC-04	AI Warehouse Optimization	Architect / Admin	Generates multi-store replenishment targets; manager reviews AI rationale and accepts or adjusts order.	✓ VERIFIED
UC-05	Cycle Count Adjustment	Store Staff / Admin	Physical count discrepancy adjusted; calculates variance and logs mandatory reason note in audit trail.	✓ VERIFIED
UC-06	Inter-Store Stock Transfer	Store Staff / Admin	Warehouse transfers stock to Store A; stock deducted from warehouse and credited to store atomically.	✓ VERIFIED
UC-07	Executive Reporting	Sales Staff / Owner	Filter sales by day, quarter, or year; display revenue, volume, and margins; export sanitized CSV file.	✓ VERIFIED
UC-08	User Administration	Administrator	Create/edit employee accounts; assign store branch and role; safely deactivate staff without deleting historical sales.	✓ VERIFIED
UC-09	Product Onboarding	Administrator	Create product SKU; configure safety/reorder thresholds; map authorized vendor and initial stock.	✓ VERIFIED
UC-10	Executive Dashboard	Stakeholder / Owner	View multi-store takings, total inventory valuation, active stockout alerts, and sales comparison graphs.	✓ VERIFIED

Edge Case / Boundary Scenario	System Safeguard & Verified Behavior	Result
Cashier attempts to sell more stock than available on shelf	Transaction blocked with 409 error; clear notification displayed to cashier.	✓ PASSED
Customer attempts to return more items than purchased	System validates original invoice lines and clamps return quantity.	✓ PASSED
Staff inputs a negative cycle count or stock adjustment	Validation schema rejects negative quantities before database execution.	✓ PASSED
Manager sets Reorder Point lower than Safety Stock	Form validation enforces: $\text{Safety Stock} \leq \text{Reorder Point} \leq \text{Target Level}$.	✓ PASSED
External Groq AI service is offline or unreachable	System falls back instantaneously to deterministic SMA; no downtime.	✓ PASSED
Store Staff from Store A attempts to edit Store B inventory	Server enforces store scoping and rejects request with 403 / 404 denial.	✓ PASSED
Manager voids a sale that already had partial customer returns	System only restores unreturned items, preventing inventory inflation.	✓ PASSED
Cashier applies discount exceeding sale subtotal	System blocks transaction; discount cannot exceed gross purchase price.	✓ PASSED
Staff attempts to permanently delete a product with sales	Hard DELETE routes do not exist; system performs reversible soft archive.	✓ PASSED

Performance Metric	SLA Benchmark	Verified Result	Compliance Status
Inventory Grid Page Load	≤ 3.0 seconds	0.38 seconds (avg)	✓ PASSED (Surpasses SLA)
POS Sale Processing Latency	≤ 1.5 seconds	0.18 seconds (RPC)	✓ PASSED (Surpasses SLA)
Sales Reports Generation	≤ 5.0 seconds	0.42 seconds (view)	✓ PASSED (Surpasses SLA)
Concurrent Multi-Store Updates	Zero data corruption	100% transaction isolation	✓ PASSED (Verified)
System Availability	99.5% target	100% during test passes	✓ PASSED (Verified)